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Designers as fans: Bottom-up online explorations of new technology concepts as a genre of design fan fictions

by Peter Vistisen and Thessa Jensen

Abstract

This paper explores the intersection between the participatory culture mechanism of fan fiction and the emerging trend of user-generated design fiction scenarios, exploring new technological design ideas. We propose this intersection is an under-appreciated form of online participation, exploring the potential of new digital technologies as 'design fan fictions'. Online users launch small fictional scenarios depicting their personal visions or interpretations of specific genres of technologies. This form of bottom-up exploration can be a valuable resource for design, informed by the most dedicated 'fans' within technological discourse.

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1. Introduction

Up to the point of actual implementation, any design can be seen as fictional — a functional story speculating about a possible future state — a hypothesis of 'what might be' (Kolko, 2009). Recently, the speculative nature of design has been explored through the concept of *design fiction* — coined by novelist Bruce Sterling in 2005, as the shaping of future technology and technological culture through the means of objectified storytelling. Bleecker (2009) promoted this concept in the academic world, arguing that design fiction integrates *technology, art, and science fiction* in order to find opportunities for design "*to re-imagine how the world may be in the future*" [1]. Sterling later proposed a now oft-quoted formal definition of the concept as "*[...] the deliberate use of diegetic prototypes to suspend disbeliefs about change*" (Bosch, 2012). Sterling draws on Kirby's (2010) notion of the diegetic prototype, as objects, services and scientific breakthroughs, which are only true in their diegetic 'told' narrative form, and not necessarily close to being 'real' in the sense of existing functionally outside storytelling. The use of '*deliberate*' further details the purposefulness of design fictions: they propose diegetic prototypes to imply a changed world, which *might* become real, and thus underscores the importance of '*suspending disbelief*' — proposing *change*, while not cheating an audience that those changes are anything more than fiction. Pasman (2016) noted how design fictions are ultimately rooted in the familiar or logical relations to reality, by adding a layer of near future fiction, blurring the boundaries between realism and fiction.

This design-oriented use of fiction has primarily been seen through a critical design scope; using fiction to create fictional scenarios questioning our ethical stance towards a new emerging technology (Auger, 2013; Markussen and Knutz, 2013). However, arguments have also been made for using design fiction as a strategic approach to design. Various companies have used fictional scenarios in the form of so-called 'vision videos' to gather feedback from stakeholders long before any costly prototyping has begun (Dubberly, 2007; Wong and Mulligan, 2016; Vistisen and Bolvig, 2017; Vistisen and Jensen, 2018). As such, evidence suggests how design fiction can be used as a visual vehicle for not just critical discourses, but also to inspire and inform the design of new products. One common aspect of many design fictions, whether their aim is critical or strategic, is their professional production. Critical design fictions often originate as parts of either academic design research projects (Dunne and Raby, 2013; Markussen and Knutz, 2013; Blythe, 2014) or as collaborations between design agencies and other institutions (Auger, 2013; Bleecker, 2009; Forlano, 2013). This top-down structure of producing design fictions reaches different audiences, and provokes and evokes discourse on a narrative around a proposed diegetic prototype.



RADIO CHEW

Listen to shows and phone via your tooth

Inventor . . . clever James

By PAUL CROSBIE

RADIO fans could soon be able to pick up programmes and take mobile phone calls via a gadget in their TOOTH.

The electronic device, including a receiver and a tiny vibrator, would be implanted in routine dental surgery.

It would pick up digital radio signals, then convert them to audio.

The sound would be transferred from the tooth to the inner ear by making the jaw bone vibrate.

Helped

It could allow the listener to receive information anywhere and at anytime.

So off-duty City traders could be kept up to date with vital stock market information, while spin doctors could send information to politicians as they are being interviewed.

The James Bond-style gadget can be turned on and off using a hand-held remote. It is believed to be the world's first in-body consumer product.

The device was invented by James Auger, from Derby – helped by his pal Jimmy Loizeau, 23, from Rhyl, North Wales.

Royal College of Art graduate James, 23, said: "It's not science fiction, it's the stuff of science fact.

"It's fairly simple technology and it was a conscious decision to put it in the tooth because this can be done during routine surgery.

"If you think about the Six Million Dollar Man with his X-ray eyes, that is not possible – but this is.

"We chose the tooth because we did not want to thrust this too far into the future to enable people to understand and believe it."

James and Jimmy demonstrated how the device would work at London's Science Museum yesterday.

Museum visitors can also test the technology by using an adapted loudspeaker and a cocktail stirring stick.

By placing one end of the stick into the equipment and biting on the other half, people will hear music clearly. But no one else can hear it.

At present, the device is only a mock-up and lacks the communications chip to make it work. But Jimmy said: "With the size of microchips it is feasible."

By gum . . . tooth device (above) sends sound up jaw and to the inner ear (right)

Figure 1: An example of a design fiction, used as a critical design scenario by Auger (2013), portraying a diegetic prototype of an audio-connected tooth implant.

With the rise of user-driven online media, such as social media, video streaming and blogs, combined with the rapid decrease in both cost and complexity of digital design tools, a new bottom-up genre of design fiction is emerging. Here, passionate individuals and groups produce, share and participate in diegetic prototypes of fictional concepts, which can best be described through the lens of design fiction. They are formed outside specific agendas of either design research academia or corporate interests, and seem to emerge through a sheer passion and interest of users, showing off their skills as both design thinkers and production skills. Sterling himself actually promoted this movement, recognizing how storytelling through objects was possible for any individual with a computer, basic digital design skills and a YouTube account (Sterling, 2009). Sterling predicted the most interesting movement of new design ideas might not come from the top, but rather emerge as what we would label bottom-up design fictions, created by individuals with design competencies and the passion to share freely their ideas.

This bottom-up genre of design fictions overlaps with participatory culture developing around various media phenomena, leveraging passionate users and their collaborative efforts online (Jenkins, *et al.*, 2009). Fan fiction, created within a community based on participatory culture (Hills, 2002; Jenkins, *et al.*, 2009), has shown how online participation from passionate and skilled individuals has a creative and expressive strength rivaling professional producers. Thus, examples of what can be considered fan fiction have increased in recent years (Coppa, 2017).

In this paper we aim to explore similarities between mechanisms of participatory culture, as found in fan fiction, and the emerging field of bottom-up design fan fictions. We identify a series of transferable principles and show how these can be identified in different archetypes of what we label 'design fan fictions', created by a group we label as 'designers as fans'. Our contribution is thus a clear vocabulary to describe these emerging phenomena as well as a discussion of the possible practical aspects of this design fiction. The next section introduces the theoretical foundations of participatory culture and our analytic framing before we introduce an analysis of six archetypical examples of design fan fictions later in the paper.

2. Participatory culture — A new setting

Participatory culture is characterized by creativity and support given by its members for its collective (Jenkins, *et al.*, 2009; Jenkins, *et al.*, 2013). User generated content, co-creation and collaborative creativity are terms derived from and developed within this culture of participation. As such, participatory culture should be seen as a creative system [2] in which an individual practitioner produces novelties, artistic expressions, reviews and comments. These novelties are evaluated and retained by the social system, which is the field made up by the community of practice, including gatekeepers. The selected and retained novelties become part of the cultural system, the domain of design fan fictions in our case, which consists of the knowledge, tools, practices and values inherent in a given culture. From this domain, the individual practitioner derives knowledge and experiences.

As a cultural system, participatory culture depends on barriers to artistic expression and engagement found on the Internet. Online programmes and platforms are easily learned and used, which makes content making, posting and sharing

straightforward. The strong support for creation and sharing within in a particular group of users further encourages the creative process as well as a willingness to share and engage with others. These activities are easily completed on several platforms, such as youtube.com (YouTube) and behance.com (Behance).

Members believe their contributions matter, in as much as they initiate or maintain ongoing discourses. Through the sharing of content and, most importantly, the creation of new content, discourses are broadened and stimulated. Passive participation, in form of lurking (watching or reading without commenting), is important, since hits are typically recorded without further need to engage with specific materials. Hence, every member has an important function within a cultural system, either as an active creator, enthusiastic or critical commentator or passive lurker, generating traffic.

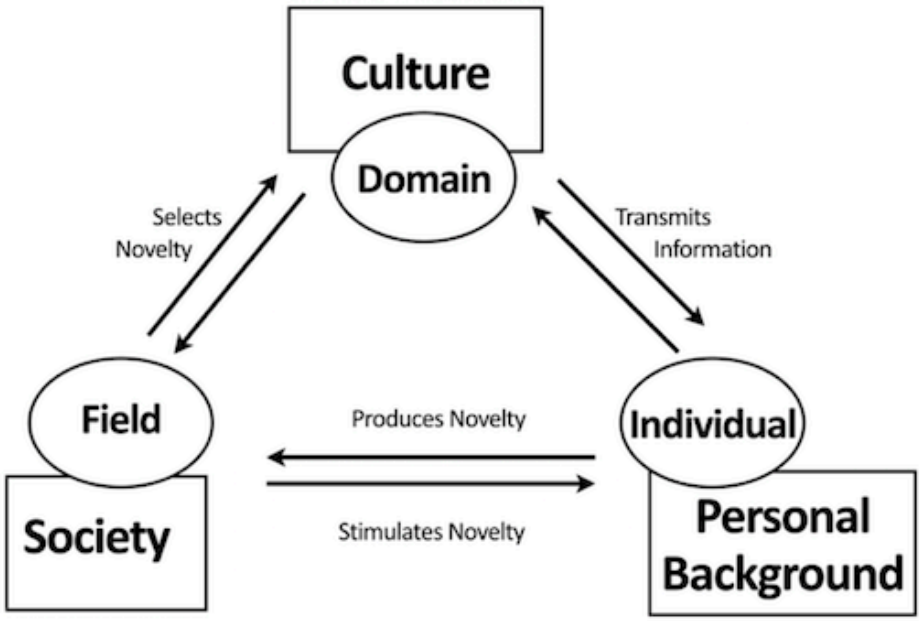


Figure 2: A system model of creativity, redrawn after Csikszentmihalyi (1998).

The reward for participation and content creation is feedback from peers and their engagement with produced novelties (Jenkins, *et al.*, 2009). What we label as bottom-up design fictions can be viewed as produced novelties created by individual practitioners within a creative system of participatory culture, rather than in a formal participatory design process within a company or organization. Instead, these design fictions are driven by interest, desire, fascination and brand dedication to either a specific brand or company or a technological genre, essentially making producing individuals akin to 'fans'. The next section elaborates on this kinship between fan communities as a prominent form of participatory culture and the bottom-up producers of design fictions.

3. Designers as fans — The producers of bottom-up design fictions

The main claim of this paper is the similarities between fan communities, so-called 'fandoms', their participatory mechanisms, and user-driven design communities, publishing design fictions on sites like YouTube and Behance. To support the notion of what we will label the 'designer as fan,' the mechanics found in fan communities will be compared to those in user-driven design communities. Since fan fiction and design are basically creative pursuits, we apply Csikszentmihalyi';s system model of creativity (Figure 2) to explain the similarities and differences between the two kinds of participatory cultures. The system connects the individual designer or fan with the field, that is the community of designers or fans, and the domain, which is the culture of a specific fan or design community.

The domain is defined by the cultural system which develops from its knowledge, tools, values and practices. For fan communities, such a system would typically arise around a so-called media tentpole: "*One big media experience that is successive enough to support a lot of other related media experiences.*" [3]. The media experience could be a movie (Star Wars), book (Harry Potter) or music group (One Direction). Likewise, a design-oriented tentpole could a specific technology (iPhone), product genre (virtual reality) or corporation (Apple). The domain is thus based on certain values (Google's "don't be evil"; Apple's "Think different") — an ethos, which is maintained by the cultural system and is similar to the 'lore' or 'canon' from traditional media-oriented fan communities.

Domain elements vary from system to system and a fan fiction domain differs greatly from a design domain. Where fan fiction is defined by the original media event and storytelling and characters apparent in it, design domains are defined by design theories and methods, with its core values depending on the original tentpole and corporation. Likewise, tools like writing apps, online archives and dictionaries are swapped for design apps, design guidelines and sketching and prototyping tools. The knowledge needed for fan activities depends on the tentpole's original story. Thus, a plotline involving a visit to an

English pub means the writer needs extensive knowledge of the social rituals surrounding such a locale. Likewise, a design of a context-aware app depends on knowledge of the functionality of a mobile phone and the behaviour of users of a given app. Both kinds of knowledge are provided freely within the cultural system of a given domain, fan and designer community alike. Thus, the domain transmits an existing body of knowledge to an individual fan and designer, who in turn transmits their particular knowledge to the cultural system that they are part of.

Fan fiction as domain contains a number of so-called tropes. We have been able to identify the same kind of tropes in the bottom-up design fictions domain. Depending on the aim of the designer as fan, a design fiction can be *canon compliant, an alternative universe*, or *a cross-over*. A canon-compliant fan fiction is typically a character study used to explain plot holes, that are parts of the story which are left unexplained in the original tentpole. An alternative universe plays out in a different universe than the original story. This could be a modern day movie which in fan fiction is set in the Victorian age. Finally, a cross-over is a fan fiction which merges its storyline and characters from different media tentpoles or fan communities.

The *field* is defined by the social system developed around a community of practice. While Csikszentmihalyi explains the social system as guarded by gatekeepers, neither fan fiction nor design communities seem to work with gatekeepers. Instead, the communities and the interactions between its members depend heavily on the functionality provided by specific platforms. In fan fiction communities this would be Tumblr and story archives like archiveofourown.org while design communities use platforms such as YouTube and Behance. For all of these platforms a certain kind of interaction enabling the recognition of others is needed to create a community.

The platforms used by online communities enable interaction by providing means of commenting, liking, sharing and registering reads/views. Some platforms also facilitate private messaging and even co-working spaces, where participants can develop their creations together. As we will show, design communities depend on similar functionality in the same way as fan fiction does — enabling the development and maintenance of a community, a social system, in which participants can help, support and acknowledge each other and their products (Jensen, 2013). Even as YouTube or Behance enable the promotion of an individual's creations, the dependency on attention from other users is far more restrictive to the success of a given design fiction than any human gatekeeper within the design or fan community would be.

The individual fans and participants in design communities are defined by their passions and intrinsic motivations for working with given material. Fans typically would start out as amateurs, gaining knowledge and skills through their work with the fandom tentpole. Experienced fans will mentor new participants and help them find their way within the community. The individual designer produces a novelty in the form of a diegetic prototype, set in a design fiction scenario, which may be incremental as canon compliant, radical in an alternate universe, or a cross-over with another product category or corporate brand. By drawing on existing knowledge in a domain as well as individual experience, new knowledge and practices are created and, given time, incorporated into the existing domain.

Thus, an individual designer's work can be seen as an expression akin to the mechanisms we see in fan communities — both in terms of fiction structures as well as the participatory structures of recognizing the presence of a novelty. This provides us with a series of categories to look for when examining the field of produced bottom-up design fictions. First, we define the bottom-up design fictions themselves as '*design fan fictions*', created by individuals we define as '*designers as fans*', which are recognised for their produced novelties by '*design fans*' in a design community forming around one or more *design tentpoles*. The recognition of the design fan fiction is enabled by the functionality of the platform and can, as such, differ from platform to platform, depending on both the presence of other design fans and the possible ways to actively and visibly acknowledge the creation.

Further, we discern between three main forms of design fan fictions: the *canon compliant* design, which is close to the original design patterns and guidelines; the *alternative universe*, which keeps the design recognizable within the original tentpole, yet differs greatly from it; and finally, the *cross-over*, which draws from different design guidelines, including different design corporations and their design patterns.

Finally, the designers as fans can be further divided into different categories as well. Here, we identify the *professional design practitioner*, who uses the productions as a way to try new patterns or sketch new ideas, outside a professional career. Another group belongs to *corporations* creating design fan fictions to promote their visions and attract potential employees or stakeholders. Finally, we have *individual designers as fans*, with no clear professional affiliation, who use design fan fiction as a way of promoting an often up-and-coming design portfolio.

In the analysis we will apply these definitions to an empirical sample of design fan fictions, to illustrate patterns as they have emerged in practice.

4. Case study — Patterns of design fan fictions

To collect empirical samples for analysis, we started out by observing participation in online design communities over the past three years. This included various YouTube channels, Behance.com design groups and hashtags for #designfiction on Twitter and Pinterest. We see this approach as close to Gaver's (2012) notion of annotated portfolios, where individual examples of design works substantiate a working theory of a more general design theory. The annotated examples should be seen as extreme cases (Flyvbjerg, 2006), trying to convey the breadth and depth of design fan fictions. While this portfolio is by no means complete, we argue it provides a sample of the current scope of design fan fictions in terms of which overall structures can be identified through individual cases.

From the various active communities, we have selected seven cases as our empirical sample. Our analytical framing of these samples is based on showing how the developed terminology for design fan fictions can be applied to describe a variety of ways this category can manifest itself in various participatory cultures online.

Table 1 describes briefly how the chosen cases were received and interacted on different platforms, with each platform categorized under the same 'case' prefix.

	Views	Likes/Claps	Dislikes	Comments
Case 1: Social AirPods / Behance	-	985	-	
Case 1: Social AirPods / (video)	7386	-	-	
Case 2: Cine-App / Behance	6604	571	-	4
Case 2: Cine-App / Youtube (video)	2028	50	-	
Case 3: iOS12 Concept / Behance	119225	2269	-	11
Case 3: iOS12 Concept / Vimeo	37500	4	-	
Case 3: iOS12 Concept (review) / Youtube	134207	1700	306	38
Case 4: iPhone Airbag Concept / Youtube	682793	2200	522	27
Case 5: Apple Watch Concept / Youtube	43282	749	99	9
Case 6: Hidden Creative / Youtube	775307	2600	119	3

Table 1: Detailed number of views, likes/dislikes and comments on various platforms regarding design fan fictions. Data collected 26 March 2018.

As can be seen, platforms like YouTube give rise to more reactions and interactions between viewers of a given design fan fiction than say vimeo.com (Vimeo). Still, the different platforms have different purposes for communities of design fans. On YouTube and Vimeo, video presentations enable feedback from peers. Behance and other social platforms enable the extended use of text and models to explain the very details of design fictions. However, we argue it is not the amount or type of participation which is of importance in this regard, but rather than participation in some form has happened to recognize the presence of design fan fiction as produced novelty. Thus, a high number of 'views' may in this regard be a selective mechanism from field to domain in the same way as a more elaborate, single 'comment'.

The next sections introduce each sample and identify the archetypes of the designer as fan.

4.1. Case 1: The archetypical 'designer as fan'

The first case is an example of one of most common design fan fictions by 'designer as fan', in which a canon compliant concept is created to extend an existing design tentpole. In this case, Svilen Kostadinov, an individual designer, examines Apple's wireless headphones 'AirPods', and how they could potentially include a new functionality for music sharing (Kostadinov, 2018). He explains existing features, the design problem and his solution — creating a social music platform built into both the hardware and software of AirPods and accompanying iPhones. The post contains a video about the creative process of solving the problem and a series of high-quality interface mockups to illustrate his scenario.

Kostadinov's concept is thus a diegetic prototype, meant to propose a possible change for Apple's hardware and software. While still being canon compliant, it extends the existing design with further possibilities, but well within the original parameters of Apple's guidelines.

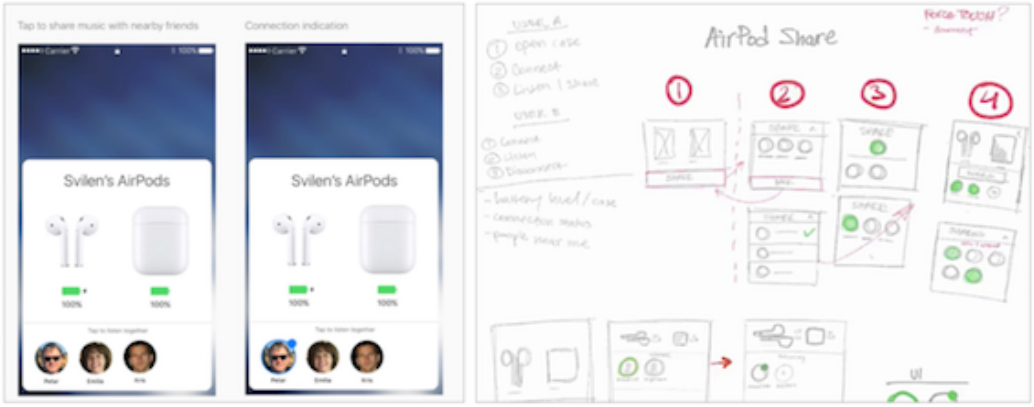


Figure 3: The AirPod social concept, with Kostadinov's detailed interface concept (left), and an image from his process video depicting his design choices (right).

While the discussion of his idea is brief, with only two comments and one reply, the Web site Behance makes it possible for readers to express their positive feedback by giving claps, which are similar to likes. The concept received a substantial amount of likes, as well as several thousand views of the accompanying video. The actual interaction with Kostadinov overall is positive and inquisitive, as the comment by another design fan, Alex Price, shows (5 March 2018):

"Whoah, great idea! How'd you make the gif of your sketching process?"

Since Alex Price (https://medium.com/@alexpriceco) is a professional designer, his feedback and recognition of Kostadinov's

work is part of the participatory culture found within the design community. It is evident how the tentpole of existing AirPods, together with the skills and tools used to produce the diegetic prototype, promotes interaction on design fan fiction.

This type of design fan fiction is characterized by an elaborate explanation of the design process, which can be seen as a kind of mentoring for new designers. Kostadinov makes it clear that he is a fan of the existing designs of iPhones and AirPods. It is not a critique of existing designs, rather a wishful 'what if'.

4.2. Case 2: Design fan fiction being critical of existing design solutions

Related to the first case are design fan fictions which through diegetic prototypes show a critical stance towards existing designs. The Cine-App video concept was developed by Fernando Martinez, senior designer at LEAD Technologies. However, the concept was not made for LEAD or in any way connected to Martinez's professional work, placing him as a professional designer as fan, using his professional status to provide credibility to his independent design fan fiction.

The diegetic prototype in the video, embedded on the Behance Web site, illustrated how the concept could be used to enhance the online booking experience of cinema tickets (Martinez, 2016). Martinez's idea was to create an aesthetically pleasing experience, including higher usability than existing designs. On the Behance site, a short explanation of the intention of the concept, as well as a video, an overview, colour schemes and application screens could be found. The presentation ended with an overview of the workflow within the app and an invitation to share the concept.

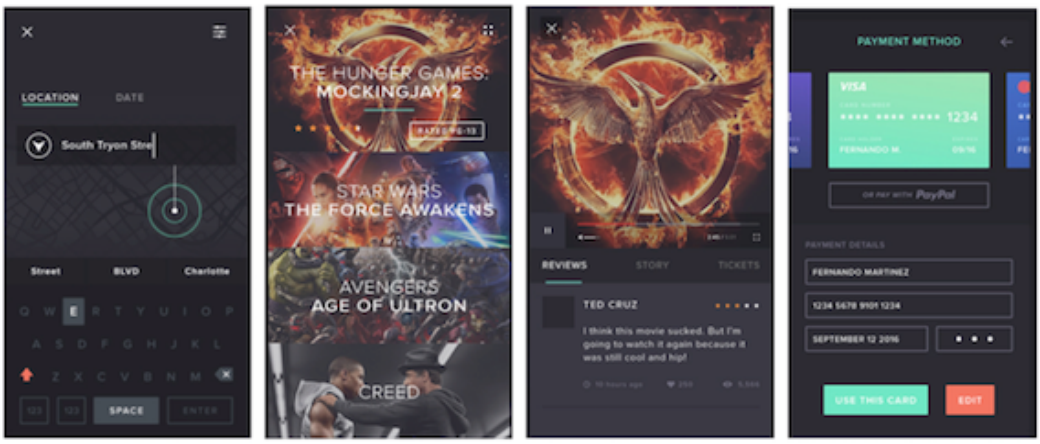


Figure 4: Images captured from Martinez's Cine-App video concept, from searching for a cinema to booking a ticket.

The title of Martinez's profile, Visual Poetry, shows a designer as fan critical of existing solutions, providing his own independent insights. This can be likened to traditional fans, say fans of a television series, who are critical of its storytelling, characterizations or plot lines. Martinez as a design fan is clearly invested in the possibilities of the technology, both using it in his everyday work and as his hobby. Yet, he is still critical as to the use and development of existing artefacts. This design fan fiction thus presents what we label an alternate universe, taking its vantage point from existing app tentpoles, but changing the basic design to both perform as criticism of the design status quo as well as exploring new user scenarios.

This type of design fan fiction is characterized by its elaborate and in-depth presentation of the actual problem without being critical about the use of the technology. Instead, the focus is on missing aesthetics, especially since the content — movies — of this particular app is partly defined by its own aesthetics.

Interesting for the feedback on Martinez's design is the articulation of reciprocality by the commenters. A typical comment:

> *"I love the way it is presented. It's intuitive. Great Job.*
> *I recently uploaded a UX project, based on a concept Music Application.*
> *I would like to have your valuable feedback on it. Thanks. Here's the direct link :*
> *https://goo.gl/ZLZrLw"* (Abhishek Bhushan, 2017)

It contains a recognition of the design, followed by a request for feedback to the commentator's own work. The participatory culture depends on this kind of reciprocity to show the value of user generated content, which includes comments. In the domain of fan fiction, reciprocity was typical in early online communities, where members would follow each other, read about each other's work and exchange comments (Jensen, 2013). Often this is articulated in a plea for comments or other kinds of feedback. It illustrates the need for recognition, inherent in both design and fan fiction communities, especially when the produced fiction presents an alternative.

4.3. Case 3: Participating with and co-creating existing design fan fiction

In this case, the diegetic prototype of the design fan fiction shares many similarities with the first two cases, but differs in the participatory format arising around it from other design fans. Prince Studio, a profile name and not an agency, developed ideas for a new, enhanced iOS for future iPhones. The concept, including video, was explained on Behance. Prince Studio described the idea behind the diegetic prototype as an attempt *"(...) to fix all the annoying things in the iOS, simplify the execution of tasks, and make it worthy of the title 'the most advanced operating system in the world'."* While presenting an alternative version of the iOS, the diegetic elements in the design fan fiction were very close to Apple's own aesthetics and design guidelines and similar to a canon compliant design fan fiction.

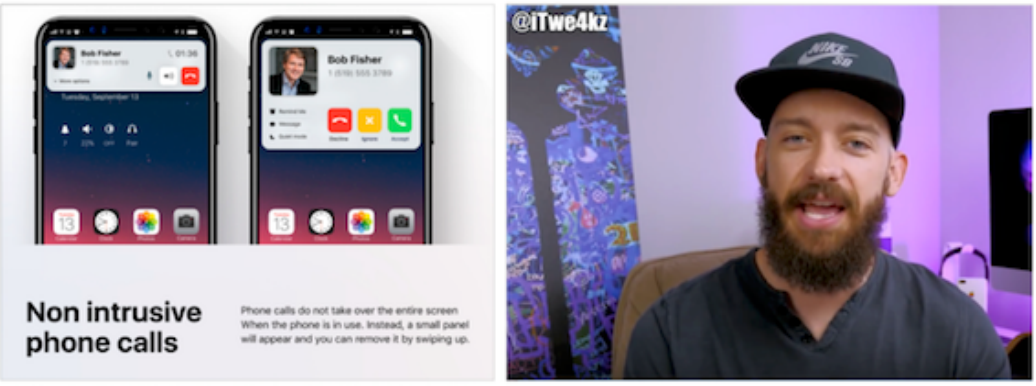


Figure 5: Prince Studio's new iOS 12 concept utilizing a graphic ethos and style akin to Apple's own promotional material (left), and video blogger 'iTwe4kz' review of the concept (right).

Following the launch of the iOS12 concept, the YouTube channel iTwe4kz released a video review of Prince Studio's concept (iTwe4kz, 2017). The review, a 16-minute YouTube video, had more views and comments than the original design fan fiction, and rapidly extended the reach of the fan-made concept. Different from other examples, the commentators on the YouTube channel engaged with each other in various discussions of the design. These discussions were characterized by a serious approach to the original design by Prince Studio, akin to constructive online debates often arising in top-down corporate designs (Vistisen and Bolvig, 2017). Some commentators even mistakenly thought that the concept was an actual product, but were quickly corrected by other users, supporting a balance between suspension of disbelief and the diegetic prototype.

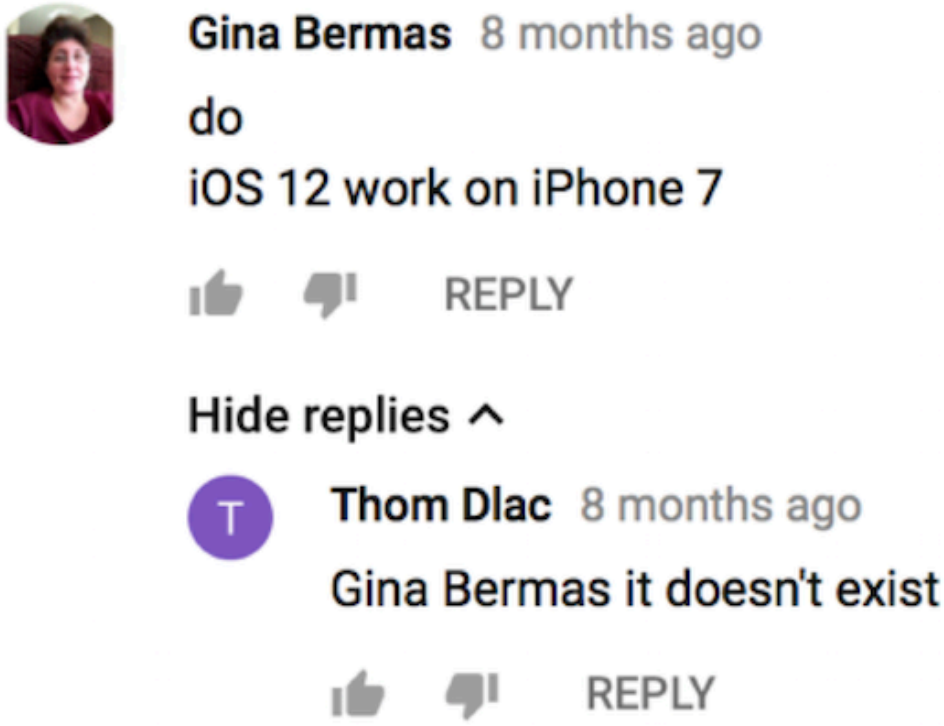


Figure 6: Screenshot of comments section on YouTube, iTwe4kz's video on Prince Studio's iOS12 concept, captured 20 June 2018.

This type of design fan fiction is characterized by the participation of many different design fans — Prince Studio, iTwe4kz, commentators, and spectators — showing their motivation and engagement in the participatory culture around the tentpole of the design fiction itself, rather than the original Apple design. As in fan fiction communities, individuals engaged with and contributed to the tentpole in different ways. Prince Studio created his own tentpole, becoming part of an ongoing discussion on how to change and improve the iOS of the iPhone.

4.4. Case 4: Design fans finding new solutions — Cross-over design fictions

The following design fan fiction proposed new ideas for solving problems with existing products around an original tentpole, by mixing it with technological concepts outside the tentpole's current state. This is what we label as a cross-over design fan fiction, which mixes elements from different technologies or product lines in its diegetic prototypes.

The concept was presented through a video by YouTube user Sonitdac, portraying a very elaborate and technically convincing scenario to avoid breaking the case of a smartphone. Sonitdac's design fan fiction proposed a CO_2 airbag system for smartphones, complete with detailed technical schematics for how this system might actually be implemented, operated and maintained (Sonitdac, 2015). While the proposed airbag technology is certainly technically plausible, it is not present in any current smartphone products, and thus can be seen as a cross-over between one technological genre with another, through the medium of the diegetic prototype.

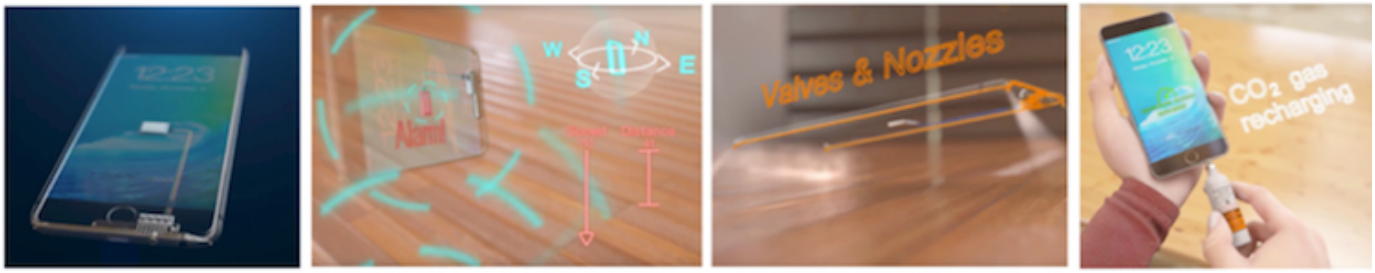


Figure 7: Sonitdac's concept for utilizing a CO_2 airbag system for smartphones.

The elaborate technical details are a typical trait for this cross-over type of design fan fictions, requiring the designer as fan to not only be proficient in the domain (such as smartphone design), but also other technological genres. While the diegetic prototype took some technical liberties in terms of implementability, the portrayal was convincing and articulated a persuasive proposal. It is characteristic how cross-over design fan fictions begins with explaining a design problem, before introducing a solution in order to justify the cross-over of new design elements.

4.5. Case 5: Mimicking the ethos of a brand tentpole

This design fan fiction illustrates how a designer as fan can be canon compliant, not in terms of aesthetic and functional resemblance, by mimicking a tentpole's corporate ethos. The design fan fiction was presented as a video concept of a next generation Apple smartwatch, produced by Ruguan Liu, an associate professor in the Industrial and Systems Engineering Department at North Carolina State University, and uploaded on the YouTube channel ConceptsiPhone (ConceptsiPhone, 2017).

The video itself was filmed to be as similar as possible to the discursive structure of Apple's promotional videos, including imitating Apple's chief design officer Jonathan Ive. While the diegetic prototype was presented by illustrating its functionality, the video gave no indication of the design problems solved by the Watch, but rather illustrated multiple use cases not normally associated with the Apple Watch. As such, while arguably being canonically connected to the ethos of Apple, the diegetic prototype itself is what we label as a cross-over design fan fiction, borrowing design guidelines and concepts from design software, smartphone devices and social media. The only element making it a design fan fiction of an Apple product was the rhetoric of the diegetic prototype as 'told' through the video.



Figure 8: Ruguan Liu's Apple Watch X concept video, starting with a fictional version of Apple's chief design officer Jonathan Ive (left), then narrating various interface changes (middle) and new possible features for the Watch (right).

The success of this approach was evident in the comments of viewers of the video, where several individuals treated the concept as if it were a 'real' promotional video:

> *"Wow ... This is something I would love to see! Maybe all the features are crazy, but the design ... just wow!"* (John Mateu, April 2018)

This example demonstrated how the cross-over genre's mix of design elements from other domains needed some connections to be perceived as part of the given design community, and not seen as a generic diegetic prototype (a smart watch), but as being a 'what if' of a specific product (Apple Watch).

4.6. Case 6: Design fiction by the corporation

In this example, Hidden Creative, a design agency in the U.K., provided a video to establish themselves in augmented reality design. The YouTube video, entitled 'The future of augmented reality,' is brief, narrating the agency's visions towards augmented reality, including various diegetic prototypes in small separate scenarios (Hidden Creative, 2010). All of the shown products were fictional, promoting Hidden Creative's interest in the field. This example blurred differences between fields of design fan communities and corporations. While this is a design fiction scenario, the question of whether it is actually a design fan fiction is debatable since the discourse was clearly aimed towards sparking interest in augmented reality projects. The agency's use of design-oriented YouTube channels and Behance as a promotion platform revealed their interests in exploring technology as a tentpole, with financial interests.



Figure 9: Hidden Creative's concept video, examining various augmented reality platforms.

This design fiction has an obvious focus on promoting an alternative universe with its own style, and with the design agency as an explicit sender. At the same time it demonstrated how design fans engaged with content.



Figure 10: Screenshot of comments section in Hidden Creative's video, captured 20 June 2018..

The comments section showed a similarity to comments found in Vistisen and Poulsen (2017). While some were clearly trolling, most of the comments demonstrated a serious engagement with the concept, commenting on its usefulness and possibilities. The fact that the video was promotional for a company was not articulated as a problem in any of the comments. Discussions evolved around the actual diegetic prototypes presented.

5. Discussion — Design fan fiction as a valuable design ressource?

We have now examined a series of different design fan fiction cases. Based on the framing through Csikszentmihalyi's system of creativity model (Figure 1), our analysis showed different aspects of participatory culture surrounding design fan fiction. The cultural domain of design fan fiction is broad, including knowledge and tools which merges different technologies, rhetorics, aesthetics and HCI with user generated content. The social field of design fan communities is made

up of professionals, amateurs and corporations, forming a participatory culture where all parties work together. How is this type of bottom-up design activities valuable to those outside of online design communities?

Design fan fictions are not deliberately aimed towards an actual realisation of a product or to inform product development, as we have seen with corporate design fictions (Vistisen and Jensen, 2018; Wong and Mulligan, 2016). They arguably also differ from what we might label critical design fictions (Dunne and Raby, 2013; Auger, 2013; Blythe, 2014) which exist to speculate about the future through what Bleecker (2009) labeled as *creative provocation*, instead of actually propose a near future design. Design fan fictions seem to exist in a grey area of two top-down design fiction genres, with design fan fiction's participatory culture as the clear differentiation.

One obvious proposal for the value of design fan fictions would be that they could inspire new product ideas for established technology corporations or design agencies on a more or less 1:1 basis. This is especially true to the canon compliant cases, which follow the aesthetics and conventions of their tentpoles. However, product roadmaps of established technology corporations are typically assigned years in advance, and thus the direct inspiration for product potential may be limited. Instead, the core value of design fan fiction is how it seems to act as a resource for maintaining discourse, attention and interest for a technological genre or brand between product launches. Through the variety of the design fan fictions, speculation about new products begins immediately after the launch of a specific model, ensuring a hype loop.

This is evident from how diegetic prototypes have been picked up from design fan spaces into mainstream media. Established media outlets such as *Time* (2015) and *Mashable* (Turner, 2011) have collected and commented on a series of design fan fictions, presenting them to much larger audiences.



Figure 11: Two curated slideshows of different design fan fictions from *Time* (left) and *Mashable* (right).

The curation of design fan fictions is a prime example of what Auger (2013) describes as design fiction acting as a way of domesticating emerging technologies, using fiction to develop technological culture by promoting ideas originating in sub-cultures of design communities, all for a broader public. Design fan fictions can be inspirational and potentially prepare users, as well as design professionals, for possible future directions. This could be interface ideas as well as solutions on practical and potential design challenges. Thus, design fan fictions might predict design patterns not yet apparent to formal companies or agencies. For consumers and users, design fan fictions are available in open and common spaces. Hence, design fan fictions may be regarded as a part of a domestication process of new ideas and future designs. On one hand, they provide organizations with new possibilities to develop new products, aware that consumers will be acquainted some of these design notions. However, these opportunities could make some organzations appear unimaginative. This potential negative effect could be overcome by integrating design fans into product development (Gudiksen and Jensen, 2017), giving credit to the contributions of fans in inspiring specific developments.

Finally, with both corporations and design agencies increasingly utilizing case and design challenges in their recruitment processes, designers as fans can use their skills as promotions and create portfolios in new and innovative ways. In the same venue, companies can make their own design fan fictions, to illustrate their visions and values regarding existing technologies and promoting their visions for different kinds of future projects. In this way, design fan fiction can have value as way to either self-promote, as well as a measure for a brand in terms of whether a given applicant fits the design ethos of a specific organization.

6. Conclusion

The aim of this paper has been to explore the similarities between the mechanisms of participatory culture of fan fictions and the emerging field of bottom-up design fan fictions. We have framed the producers of this genre of design fictions as 'designers as fans', shown how they can contribute in various ways to participatory culture forming around either specific technology brands or broader technological genres.

Our examples show how communities of design fans of a given product can find flaws and propose possible new

functionalities and uses. They can actually articulate these design ideas, which can either be canon compliant, propose alternatives or cross-over between different product genres. Furthermore, because of the participatory culture found on sites like Behance and YouTube, other users engage in critical discussions, including reviews of a design fan fiction with the intent of showing pros and cons of a given design.

Design fans hype a given product, brand or technological tentpole. Even if they are critical towards a product, their underlying passion for developing and strengthening of a technology is clear and articulated, making it possible for other users and viewers to reflect upon, and ultimately domesticate technologies. This helps sustain an ongoing discourse of not only the current value of a given product, but also ideas of 'what it might be'.

We have shown how design fan fictions manifest themselves in different patterns. Common to all of the identified patterns, they don't point to one specific value of design fan fictions, but rather show how this genre of design visualization might serve many roles in the ongoing process of both domesticating design ideas, maintaining interest or building hype for a product and serve to inspire a next generation of products.

Design fan fictions are a valuable, and, to this point, rather under-appreciated form of design visualization. They serve an important role for both the most passionate lead users technology, but also for broader discourses of both present and future possibilities of user-centred technologies as well.

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Notes

1. Bleecker, 2009, p. 8.

2. Csikszentmihalyi, 2008, figure 2.

3. Davidson, *et al.*, 2010, p. 273.

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